Band-1 Hadamard Order 3524 — Miyamoto (H_880 = H_20 H_44); Gap 2052 Still OPEN

H_20/H_44/H_880/C_882/H_3524 PASS; 3524 Miyamoto-only bonus (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_3316_Miyamoto`; `2026-09-15_Band1_Hadamard_3284_Miyamoto`; `2026-09-15_Band1_Hadamard_3188_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_3524_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 20, 44, 880, and 3524 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Kronecker seed $H_{880} = H_{20} \otimes H_{44}$. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 3524 is a **Miyamoto-only bonus** (NOT in SPK): Paley-I **FAIL** ($q = 3523$ not prime); Paley-II **FAIL** ($q = 1761$ not prime); Kronecker **FAIL** ($3524 = 4 \cdot 881$, 881 not Hadamard-admissible). Seed H_{880} uses Kronecker(H_{20} Paley-I $q = 19$, H_{44} Paley-I $q = 43$); Paley-I/II seeds for 880 **FAIL**. `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 3316 on box. This increment ships **order 3524** ($3524 = 4 \cdot 881$, 881 prime $\equiv 1 \pmod 4$). Seed H_{880} via Kronecker of Paley type I matrices H_{20} ($q = 19$) and H_{44} ($q = 43$); Paley-I seed for 880 **rejected** ($q = 879$ not prime); Paley-II seed for 880 **rejected** ($q = 439$ prime but $\equiv 3 \pmod 4$). Symmetric Paley conference C_{882} over prime 881. $H_{3524} = \text{Miyamoto}(H_{880}, C_{882})$. Unit tests **28/28 PASS**. SPK next gap remains **2052** (OPEN). Order 3524 is **Miyamoto-only** (NOT in SPK). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 3524; Miyamoto; Kronecker; Paley type I; Paley conference; constructive gap 2052; SPK; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{20} constructed and verified (Paley type I, $q = 19$). 2. H_{44} constructed and verified (Paley type I, $q = 43$). 3. H_{880} constructed and verified (Kronecker $H_{20} \otimes H_{44}$). 4. C_{882} constructed and verified (Paley conference, $q = 881$): $CC^\top = 881I$, zero diagonal. 5. H_{3524} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 6. SPK: $3524 \notin$ SPK (Miyamoto-only bonus). Paley-I/II/Kronecker paths for 3524 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$3524 = 4 \cdot 881$
$q = 881$	prime, $881 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{880}
Needed conference	order $882 = q + 1$
H_{880} path (used)	Kronecker(H_{20} Paley-I $q = 19$, H_{44} Paley-I $q = 43$)
H_{880} Paley-I	**REJECTED**: $q' = 879$ not prime
H_{880} Paley-II	**REJECTED**: $2(q' + 1) = 880 \Rightarrow q' = 439$ prime but $\equiv 3 \pmod 4$
Conference path	Symmetric Paley over prime 881
Paley-I for 3524	**REJECTED**: $q' = 3523$ not prime
Paley-II for 3524	**REJECTED**: $q' = 1761$ not prime
Kronecker for 3524	**REJECTED**: only 4×881; $881 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{20}	Paley type I, $q = 19$	0	0.0002	**PASS**
H_{44}	Paley type I, $q = 43$	0	0.0008	**PASS**
H_{880}	Kronecker(H_{20}, H_{44})	0	0.3036	**PASS**
C_{882}	Paley conference, $q = 881$	0	0.6719	**PASS**
H_{3524}	Miyamoto(H_{880}, C_{882})	0	19.1315	**PASS**

Primary pipeline wall (incl. unit tests + SPK refresh) ≈ 20.8449 s. Compressed ± 1 matrices saved as `H20_pm1.npz`, `H44_pm1.npz`, `H880_pm1.npz`, `C882_pm1.npz`, `H3524_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible ≤ 2000 $\cup$ on-box prior ships (through 3316/828/820/…) $\cup$ verified H_{880} $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK**; order 3524 does **not** enter SPK.

Quantity	Value
Known count	886
3524 in SPK	**False** (Miyamoto-only bonus)
880 in SPK	**True** (via Kronecker / ≤ 2000)
Next constructive gap	**2052** (OPEN)
Miyamoto-only bonus?	**Yes** (like 3284/3076/3044)

5. Gap 2052 (OPEN; method triage / non-finding ONLY)

Path	Result
Paley-I ($q = 2051$)	FAIL
Paley-II ($q = 1025$)	FAIL
Kronecker	FAIL ($2052 = 2^2 \cdot 3^3 \cdot 19$)
Miyamoto + Paley conference ($q = 513$)	FAIL (513 not prime)
GS / SDS / Williamson / Yamada	lit-only; not constructed this run

Status: OPEN. Non-finding $\neq$ counterexample. Conjecture remains OPEN.

6. Unit tests

28/28 PASS (Sylvester/Paley/Miyamoto controls; $q = 19/43/881$ arithmetic; H_{880} Kronecker path; Paley seed paths for 880 rejected; SPK path rejection for 3524; reject-all-ones; conference control; gap-2052-not-filled guard).

7. Non-claims (explicit)

- No full Hadamard conjecture claim.
- No Clay-closed stamp; `clay_closed = false`.
- No finished-ToE claim; GATE-04 **LOCKED**; no invented J.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.
- Gap **2052** remains **OPEN** under SPK.
- Order 3524 is **Miyamoto-only** (NOT folded into SPK).

8. Next optional

method-attack SPK gap **2052** OR next Miyamoto after 3524 (triage seedable $q > 881$, e.g. $q = 929 \to n = 3716$ / $q = 937 \to n = 3748$) OR Track B privacy / Object 1 meter await.

Lab: `/workspace/experiments/band1/hadamard_3524_miyamoto/` · *Script:* `verify_hadamard_3524.py` · *Artifacts:* `results.json`, `SUMMARY.txt`, `literature_ledger.json`, `*.npz`.